COVID-19 AI Risk Monitoring System

An End-to-End Databricks Data & AI Pipeline

Demonstrating the design and implementation of a scalable COVID-19 risk monitoring system using Databricks. This integrated solution combines data engineering, machine learning, automation, and analytics to transform raw health data into AI-driven decision-support insights for public health teams.

Presented by: Ruchi Wange

Acknowledgements: Databricks, Codebasics, Indian Data Club

Problem Statement & Objective

The Challenge

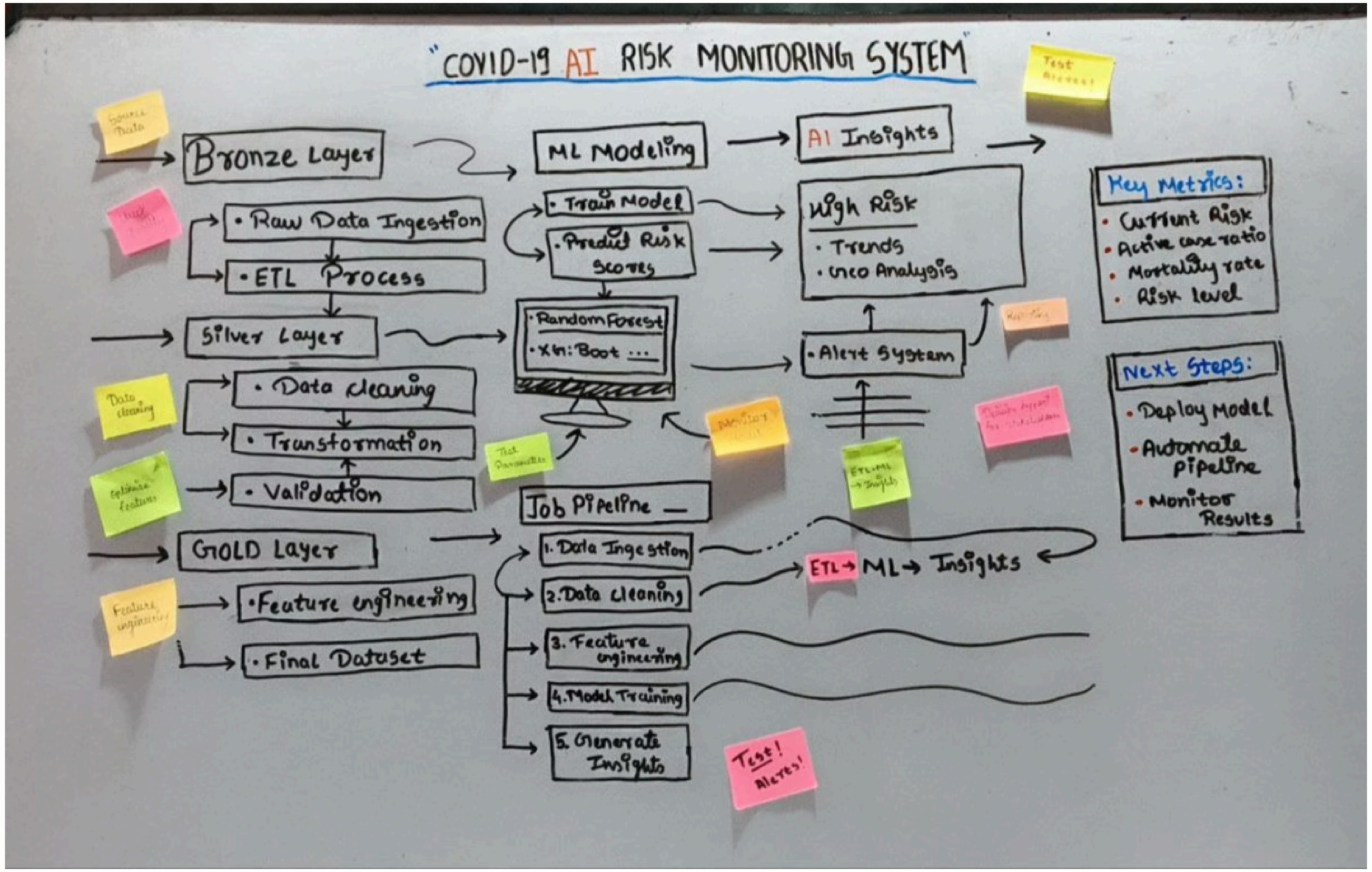
Public health decision-making requires timely, interpretable insights rather than raw data streams. Static reports fail to capture evolving risk patterns in real-time, creating gaps between data availability and actionable intelligence.

Project Objective

Build a scalable, automated system that monitors COVID trends, predicts risk levels using AI, and presents insights in a decision-ready format for health data teams.

Planning & System Architecture

Before implementation, the complete workflow was planned on a whiteboard to ensure clarity, scalability, and real-world alignment. This planning-first approach reduced complexity and ensured clean execution.



This whiteboard represents the initial planning phase where the end-to-end workflow was designed before coding.

It helped define clear data layers, ML integration, and decision-support outcomes, ensuring a structured and scalable implementation.

Data Pipeline: Medallion Architecture

The data pipeline follows the Medallion Architecture pattern, ensuring reliability, traceability, and scalability across three distinct layers.

Bronze Layer

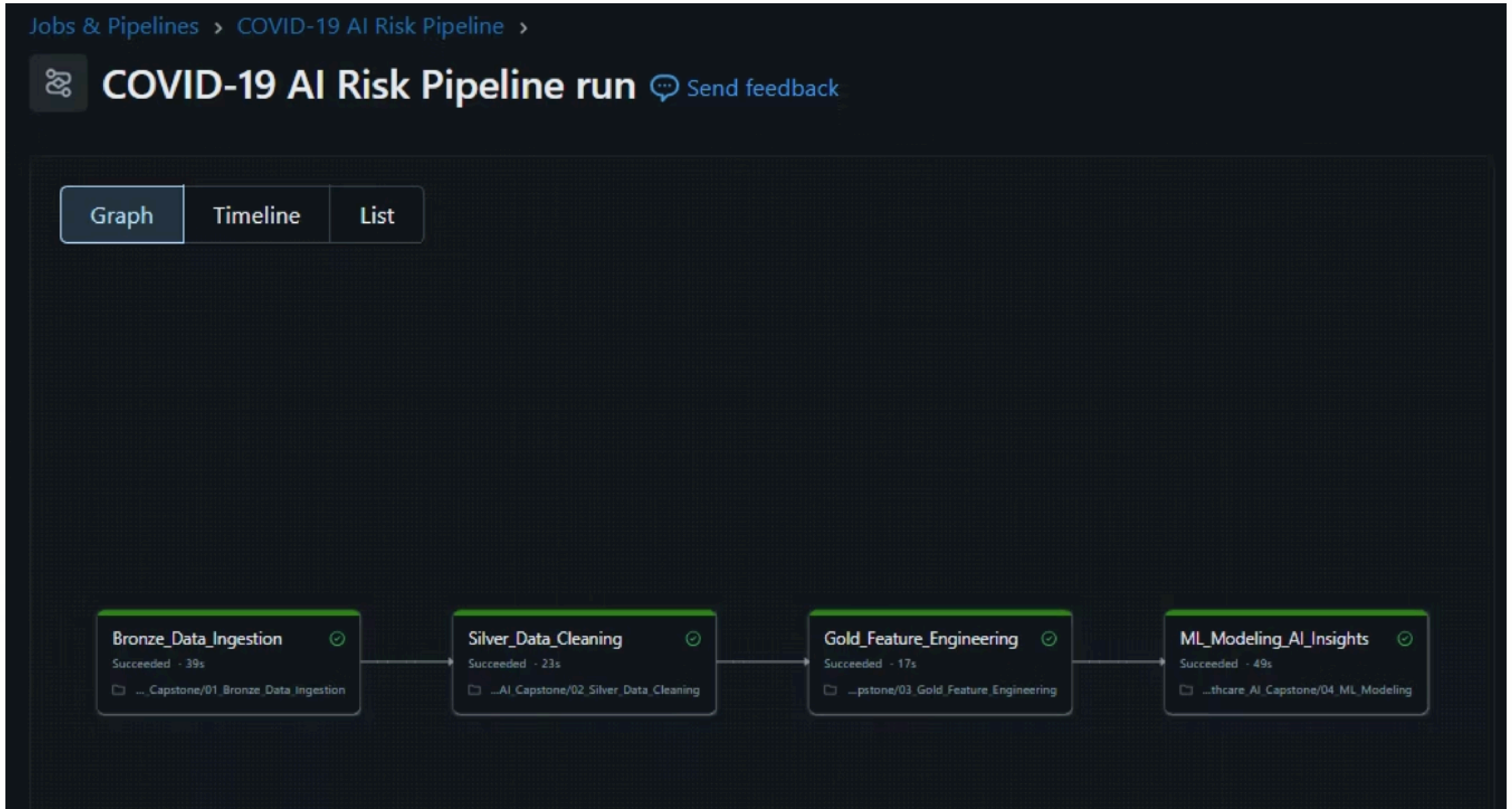
Raw ingestion of COVID case and vaccination data from source systems

Silver Layer

Data cleaning, standardization, and quality handling operations

Gold Layer

Feature engineering for analytics and AI readiness



- Databricks Jobs were used to automate the complete end-to-end pipeline, orchestrating data ingestion, transformation, feature engineering, and machine learning in a consistent and repeatable workflow.
- The Bronze, Silver, Gold, and ML stages are configured as dependent tasks, ensuring correct execution order and mirroring real-world production pipelines.
- A successful job run validates pipeline automation, reproducibility, and readiness for scalable execution beyond manual notebook runs.

Feature Engineering & AI Readiness

Engineered Features

Instead of relying on raw counts, domain-driven features were engineered to capture trends and severity:

- Growth rates and acceleration patterns
- Case-to-vaccination ratios
- Mortality indicators and severity scores
- Temporal trend analysis

These features enable machine learning models to learn meaningful patterns and produce interpretable outputs.

Machine Learning & AI Insights

01

Model Training

ML model trained using Gold-layer features to classify COVID risk levels

02

Interpretable Outputs

Model produces interpretable risk predictions for human review

03

Delta Storage

Insights stored as Delta tables for downstream consumption

04

Actionable Intelligence

Transforms analytical signals into actionable AI insights

Pipeline Automation with Databricks Jobs

The entire pipeline—from data ingestion to AI insights—is automated using Databricks Jobs. Each stage is orchestrated as a dependent task, enabling reproducible and production-style execution with a single trigger.

Bronze Stage

Raw data ingestion and storage

Silver Stage

Data cleaning and quality control

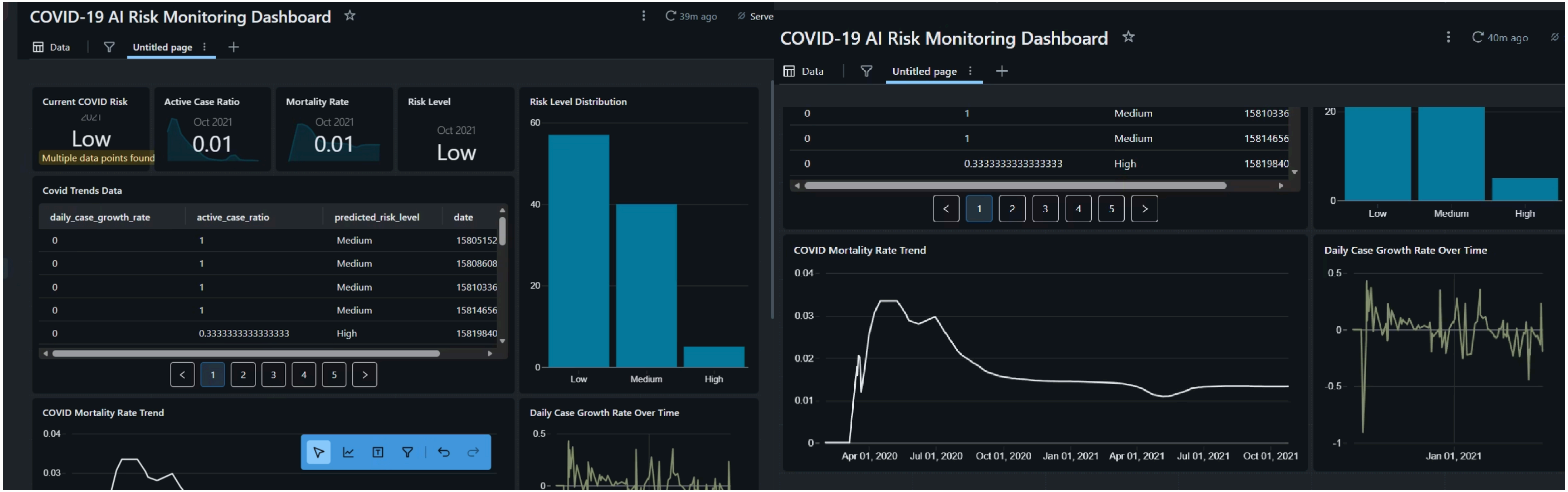
Gold Stage

Feature engineering and preparation

ML Stage

Model inference and risk prediction

Analytics & SQL Dashboard



📄 Why a Single-Page Dashboard

- A single-page dashboard was intentionally designed to present all critical insights in one consolidated view, avoiding fragmentation and ensuring stakeholders can understand the current risk situation without navigating across multiple pages.

📊 Dashboard Explanation

- The dashboard provides an executive-level view of COVID-19 risk by combining AI-predicted risk levels with key health indicators such as active case ratio, mortality rate, and daily case growth trends, enabling quick and informed decision-making.

Business Impact & Decision Support

Bridging the Gap

Connects data, AI, and decision-making through interpretable risk insights

Faster Assessment

Enables rapid evaluation of public health risk levels across regions

Proactive Monitoring

Supports early detection of emerging risk patterns

Operational AI

Demonstrates real-world impact of operationalized machine learning

THANK YOU

Presented by: Ruchi Wange